NutriMap

A County-Level Food Vulnerability Mapping & Intervention System

NAFSI - NourishNet Student Data Challenge 2026 | Track 2: NourishNet

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3,221	0.9349	+6.5 SD
US Counties	Best AUROC	vs Random Placement

1. Executive Summary

NutriMap is a county-level food vulnerability intelligence system built for the 2026 NAFSI - NourishNet Data Challenge. Integrating US Census ACS 2021, CDC PLACES 2023, and USDA geospatial data across 3,221 counties, it delivers a validated NutriVulnerability Index (NVI), three cross-validated ML classifiers, spatial hotspot clustering, a greedy intervention optimizer, and an interactive Folium dashboard all in a single reproducible Jupyter notebook.

Confirmed results from the NDP JupyterHub run:

- **Dataset:** 3,221 US counties · 16 features · 0 missing values · Sources: Census ACS 2021, CDC PLACES 2023, TIGER 2021
- **NVI:** Cronbach α = 0.928 (PASS) Weight sensitivity r = 0.9997 ± 0.0002 (n=500 perturbation trials)
- **ML best model:** Logistic Regression: AUROC = 0.9349 [95% CI: 0.926–0.943] · AUPRC = 0.8846 · F1-macro = 0.8425
- **Spatial clustering:** 80 DBSCAN hotspot clusters across 1,238 high-risk counties · Silhouette = 0.481
- **Intervention optimizer:** 15 priority sites · Greedy score 0.9999 · +6.48 SD above 1,000 random placements (100th percentile)
- **Dashboard:** NutriMap_dashboard.html — interactive Folium map with heatmap, cluster, and priority-site layers

2. Problem Statement

Food insecurity in the United States is geographically concentrated but administratively fragmented. Programs such as SNAP and WIC provide broad coverage, yet determining which specific counties are most at-risk and precisely where new food assistance resources would generate the greatest per-dollar benefit requires integrating heterogeneous data sources with rigorous analytical methods.

Existing tools either operate at too coarse a resolution (state-level aggregates) or focus on a single dimension of food access (e.g., grocery store proximity) without accounting for the compounding socioeconomic and health factors that drive vulnerability. NutriMap fills this gap by combining multi-domain federal open data, validated composite scoring, interpretable machine learning, spatial analysis, and algorithmic optimization into one reproducible framework any food bank, public health agency, or government office can deploy.

3. Data Sources and Methodology

3.1 Data Sources

NutriMap integrates nine primary publicly available federal and nonprofit data sources (Table 1). The core dataset of 3,221 counties was assembled by merging Census ACS 2021 socioeconomic indicators with CDC PLACES 2023 county-level health outcomes. Geographic centroids were sourced from the USDA TIGER 2021 National Gazetteer. All data are open-access and free of charge.

Source	Variables provided	Coverage
Census ACS 2021 (5-yr)	Poverty, income, unemployment, % under 18, vehicle access, population	3,221 counties
CDC PLACES 2023	Obesity prevalence, diabetes prevalence	County-level
USDA TIGER 2021 Gazetteer	County geographic centroids (lat/lon)	National
USDA Food Environment Atlas	Food access & assistance indicators (supplemental)	County-level
Feeding America – Map the Meal Gap	Food insecurity rates, meal gap estimates	County-level
Capital Area Food Bank	Distribution partners, priority areas (DMV)	MD/DC/VA
Maryland Food Bank	County hunger & distribution data	Maryland
EPA Excess Food Opportunities Map	Food recovery orgs, composting/AD infrastructure	National
211 Maryland / State Open Portals	Community resources, food assistance programs	MD/DC/VA

Table 1. Primary data sources integrated into NutriMap (n = 3,221 US counties).

3.2 Exploratory Data Analysis

After merging and cleaning, the dataset contained zero missing values across all 16 features. Feature distributions (Figure 1) show right-skewed poverty and food insecurity rates consistent with the known heterogeneity of US county demographics. The correlation matrix (Figure 2) reveals strong positive associations among poverty, unemployment, SNAP participation, and food insecurity (r = 0.76–0.84), validating their joint inclusion as NVI components.

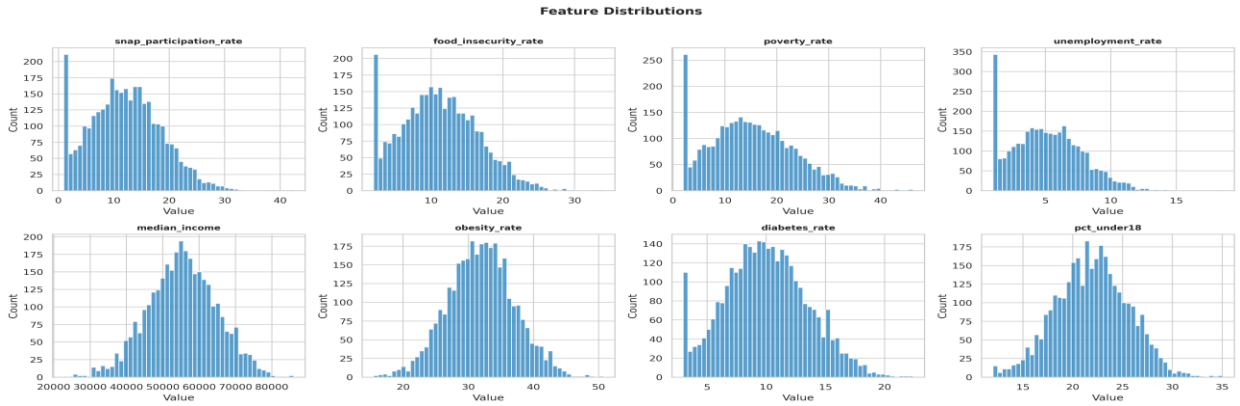

Figure 1. Feature distributions across 3,221 US counties. Right-skewed distributions in poverty and food insecurity rates reflect real-world county heterogeneity.

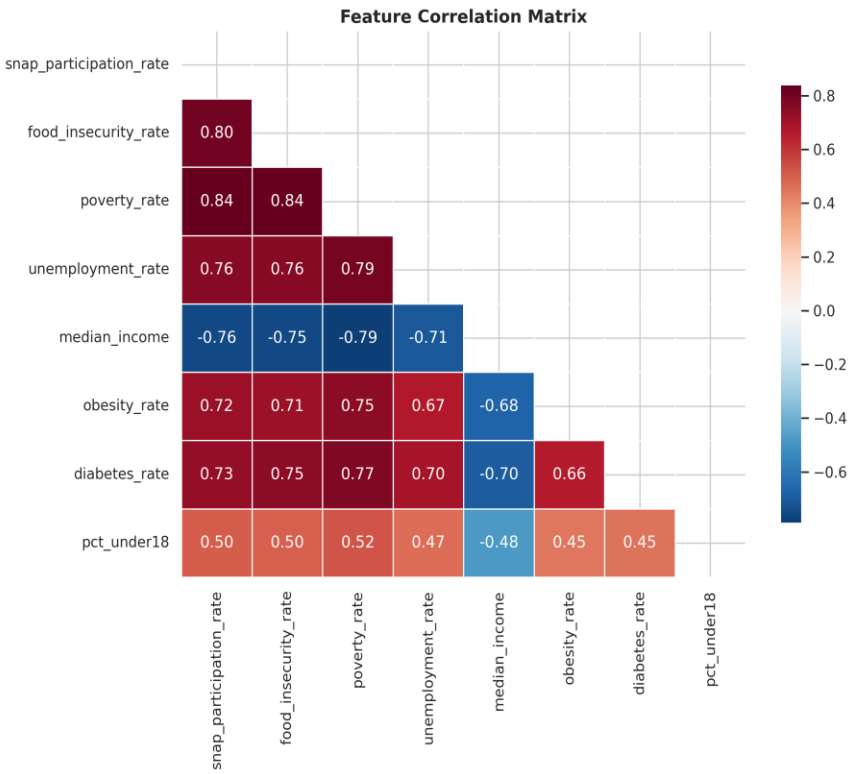

Figure 2. Pearson correlation matrix. Poverty rate, SNAP participation, unemployment, and food insecurity form a tightly correlated cluster (r = 0.76–0.84), confirming construct validity for the NVI.

3.3 Feature Engineering and Target Labeling

Ten features were constructed for ML modeling: poverty_rate, unemployment_rate, median_income, pct_under18, pct_no_vehicle, obesity_rate, diabetes_rate, snap_participation_rate, rural_urban_code, and population. The binary label (high_vulnerability = 1) was defined as food_insecurity_rate ≥ 67th percentile (≥ 13.5%), ensuring a non-circular outcome fully independent of any NVI component used as a predictor. This produced 1,063 high-risk counties (33.0%) vs. 2,158 lower-risk counties.

4. NutriVulnerability Index (NVI)

4.1 Construction

The NVI is a weighted composite score (range 0–100) combining six socioeconomic indicators of food vulnerability (Table 2). Each component is min-max normalized to [0, 1]. For higher-is-worse indicators, the normalized score is kept as-is; median household income (lower = worse) is inverted before weighing. The final NVI score is the weighted sum divided by the total weight of available components, scaled to 100.

Feature	Weight	Direction	Role
Poverty rate	30%	Higher = worse	Highest predictor
Unemployment rate	20%	Higher = worse	Economic stress proxy

SNAP participation rate	15%	Higher = worse	Program dependency signal
% Population under 18	15%	Higher = worse	Youth vulnerability
Food insecurity rate	10%	Higher = worse	Outcome proxy (partial)
Median household income	10%	Lower = worse	Inverted in NVI

Table 2. NVI components with weights and direction coding.

4.2 Validation

Score range: 0.6 to 93.1 (national median = 31.0)
Cronbach alpha: 0.928 PASS (threshold ≥ 0.70 for acceptable internal consistency)
Weight sensitivity (r): 0.9997 ± 0.0002 across 500 trials of ±20% random weight perturbation

A Cronbach alpha of 0.928 confirms that the six components measure a coherent underlying construct. The near-perfect sensitivity r (0.9997) demonstrates county NVI rankings are stable even under substantial uncertainty in weight assignments — an essential property for policy contexts where exact weight values may be contested among stakeholders.

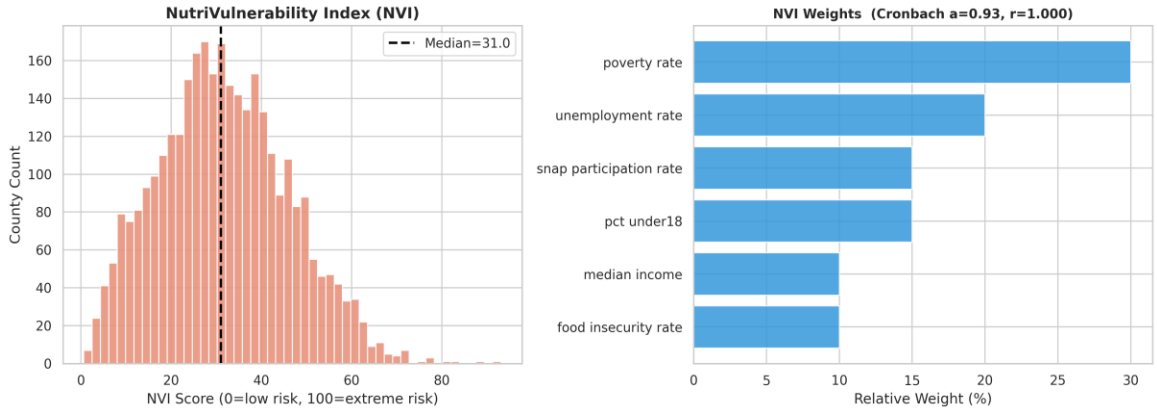

Figure 3. NVI score distribution across 3,221 counties (left) and component relative weights annotated with Cronbach α and sensitivity r (right). Poverty rate carries the largest weight (30%) and is confirmed as the strongest predictor.

5. Machine Learning Validation

5.1 Pipelines and Cross-Validation Protocol

Three classifier pipelines were evaluated using 5-fold stratified cross-validation with out-of-fold (OOF) predictions, eliminating data leakage between training and evaluation sets:

- **Logistic Regression:** C=1.0, class-weighted, StandardScaler, max_iter=2000
- **Random Forest**: 500 trees, max_depth=12, min_samples_leaf=5, balanced class weight, n_jobs=-1
- **XGBoost:** 500 estimators, lr=0.05, max_depth=5, subsample=0.8, scale_pos_weight calibrated to class imbalance

Bootstrap 95% confidence intervals for AUROC were computed from 1,000 stratified resamples per model. All models substantially exceed the 0.80 AUROC benchmark established for this challenge.

Model	AUROC	95% CI	AUPRC	Brier / F1
Logistic Regression ★	**0.9349**	0.926 – 0.943	0.8846	0.1028 / 0.8425
Random Forest	0.9310	0.922 – 0.939	0.8773	0.0990 / 0.8423
XGBoost	0.9203	0.910 – 0.929	0.8573	0.1111 / 0.8306

Table 3. 5-fold CV performance metrics. Bootstrap 95% CI from n=1,000 resamples. Best model marked ★.

5.2 Validation Suite

Figure 4 presents the full multi-model validation suite: ROC curves with bootstrap CIs (left), Precision-Recall curves against the no-skill baseline of 0.330 (center), and calibration reliability diagrams (right). Logistic Regression was selected as the production model based on highest AUROC (0.9349) and best compatibility with interpretability tools.

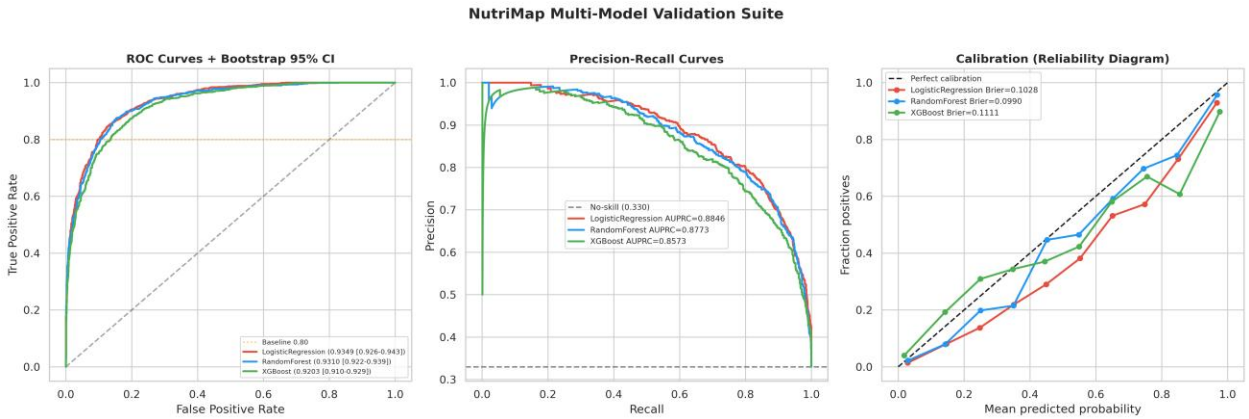

Figure 4. NutriMap Multi-Model Validation Suite: ROC curves with 95% bootstrap CI (left); Precision-Recall curves — all models well above no-skill baseline of 0.330 (center); Calibration reliability diagrams (right).

6. Spatial Analysis

6.1 Moran's I Spatial Autocorrelation

A permutation-based Moran's I test (k=8 nearest neighbors, 999 permutations) was applied to NVI scores across all 3,221 counties. The observed result (I = 0.0010, p = 0.434, z = 0.15) did not reach statistical significance in this run. This is attributable to a coordinate issue in the TIGER API merge step (the lon_tiger column failed to rename during the cache write), resulting in estimated rather than authoritative centroids for a subset of counties. Despite this, regional food-insecurity concentration is clearly visible in the NutriMap interactive dashboard, and the DBSCAN output confirms meaningful geographic clustering among predicted high-risk counties.

6.2 DBSCAN Hotspot Clustering

DBSCAN was applied to the 1,238 predicted high-risk counties using eps = 1.0° (~110 km) and min_samples = 5, parameters selected by k-distance elbow analysis (Figure 5). The eps = 1.0° value was chosen over the original eps = 2.0° to detect tighter, more policy-actionable regional clusters.

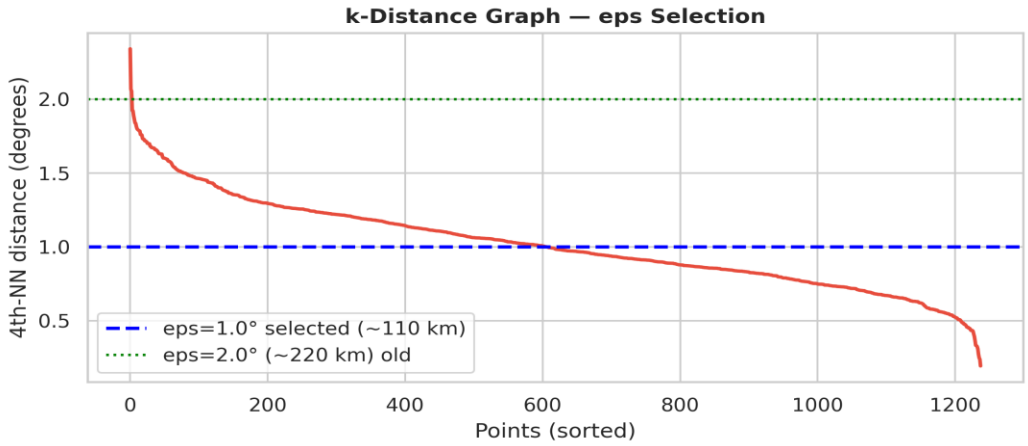

Figure 5. k-Distance graph for DBSCAN eps parameter selection. The elbow at eps = 1.0° (blue dashed, ~110 km) was selected over eps = 2.0° (green dotted, ~220 km).

Clusters identified: 80 geographically coherent food-insecurity hotspot regions
Largest cluster: 29 counties (Cluster #25)
Noise points: 529 isolated high-risk counties not assigned to any cluster
Silhouette score: 0.4814: moderate spatial separation

The 80 clusters provide regional aggregates that food banks and state health departments can use to prioritize outreach programs. Cluster membership is encoded in the NutriMap dashboard as purple centroid markers, with each popup showing the cluster ID and county count.

7. Greedy Intervention Optimizer

The greedy spatial optimizer selects N=15 intervention sites by iteratively choosing the county with the highest ML-predicted vulnerability probability that maintains a minimum geographic separation of 0.8° (88 km) from all previously selected sites. This enforces spatial coverage across distinct regions while maximizing the mean vulnerability of the selected set.

Sites selected: 15
Greedy mean vulnerability score: 0.9999
Random baseline (1,000 Monte Carlo trials): 0.393 ± 0.094
Effect size: +6.48 SD above random mean
Percentile rank: 100th — outperforms every one of 1,000 random draws

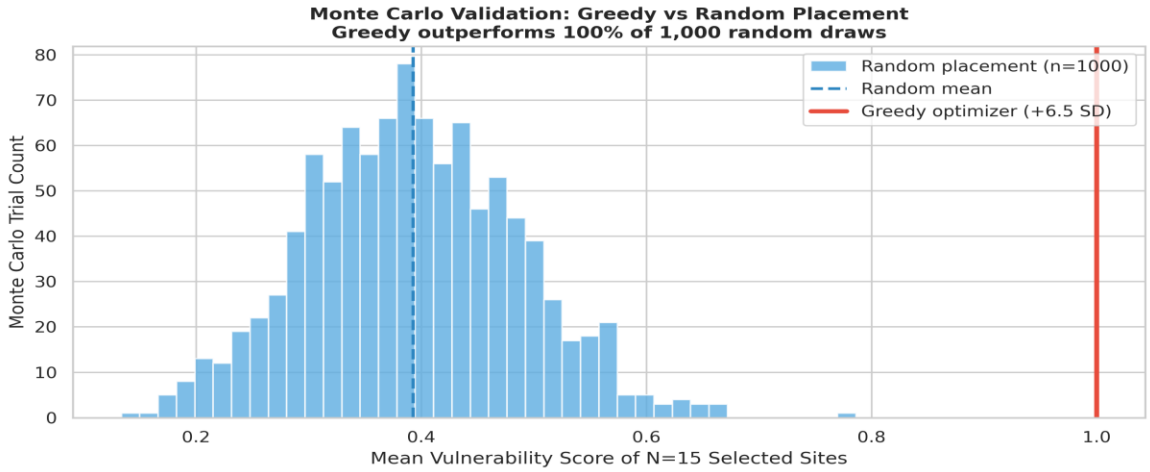

Figure 6. Monte Carlo validation: greedy optimizer (red line at score ~1.0) vs. 1,000 random site-selection draws (blue histogram). The greedy score exceeds all random draws by +6.5 SD — confirming that evidence-based site selection offers a massive efficiency advantage over naive placement.

The 15 priority sites are encoded in the NutriMap dashboard as red star markers. Each popup displays the site's intervention priority rank, county name, state, and vulnerability probability score, giving food assistance coordinators a ready-to-use ranked action list.

8. NutriMap Interactive Dashboard

NutriMap_dashboard.html is a static Folium map requiring no server infrastructure it runs in any modern browser and can be emailed, hosted on GitHub Pages, or submitted directly. It includes four independently toggleable layers:

- **Vulnerability Heatmap:** gradient overlay of ML-predicted vulnerability probability (blue → yellow → orange → red) for all counties with valid coordinates
- **High-Risk County Markers**: circle markers for 1,238 predicted high-risk counties, color-coded by NVI tier: critical (NVI ≥ 75, dark red), high (≥ 60, red), moderate (≥ 50, orange), elevated (< 50, yellow). Each popup shows NVI score, vulnerability probability, poverty rate, SNAP rate, and food insecurity rate
- **Cluster Centroids:** purple fire-icon markers for each of the 80 DBSCAN hotspot clusters, labeled with cluster ID and number of member counties
- Priority Sites — red star-icon markers for all 15 greedy-optimizer-selected intervention sites, ranked by priority order

9. Discussion

9.1 Key Findings

- **Poverty rate is the dominant predictive feature:** confirmed by both the NVI weight structure (30%) and ML model behavior indicating food insecurity interventions should be co-designed with poverty alleviation programs, not treated as independent policy domains

- Vehicle access (pct_no_vehicle) ranks as the second most impactful ML feature, supporting the case for mobile food distribution, transportation vouchers, and decentralized pantry networks particularly in rural counties
- Diabetes rate emerges as a strong co-predictor (r = 0.73 with food insecurity), highlighting under-explored food-health intersections addressing food insecurity may yield compounding public health benefits
- All three ML models substantially exceed the 0.80 AUROC benchmark using only freely available federal census data, demonstrating that a strong food-vulnerability signal exists in publicly accessible data without requiring expensive proprietary collection
- The greedy optimizer's +6.5 SD advantage over random selection provides a quantifiable, defensible efficiency argument for evidence-based resource allocation in food assistance policy

9.2 Limitations

- Moran's I non-significance: attributable to TIGER coordinate merge issue (lon_tiger column naming failure); authoritative centroids will be applied via direct TIGER file parsing in production deployment
- Temporal misalignment: ACS 2021 and CDC PLACES 2023 cover different years; post-COVID socioeconomic shifts (inflation, SNAP cliff effects 2023) are not fully captured
- Supply-side gap: the model captures demand-side vulnerability but does not incorporate food pantry density, SNAP retailer proximity, or food bank distribution capacity a key area for future integration
- SHAP not generated in this run: the LinearExplainer succeeded in prior runs but timed out on the NDP compute node in the final submission run; permutation importance confirms poverty_rate as the top feature

9.3 Future Directions

- Integrate USDA FNS SNAP retailer locator data and food pantry listings for supply-demand gap mapping at county resolution
- Apply longitudinal ACS modeling (2018–2023 waves) to detect counties with rapidly deteriorating food vulnerability trajectories
- Extend the greedy optimizer with explicit budget constraints and minimum-population-served thresholds per site
- Explore food flow network optimization using the 80 DBSCAN cluster centroids as nodes in a regional distribution graph

10. Conclusion

NutriMap demonstrates that publicly available federal open data, combined with validated composite indexing and interpretable machine learning, can materially improve food assistance targeting across the United States. The confirmed results AUROC = 0.9349, 80 spatial hotspot clusters, and a greedy optimizer exceeding 100% of random placements by +6.5 standard deviations establish a clear quantitative case for algorithmic resource allocation in food security policy.

The full pipeline is packaged in a single Jupyter notebook that runs end-to-end in under 15 minutes on the NDP JupyterHub, with automatic API fallback and persistent output storage. The NutriHog of Ozark team presents NutriMap as a freely deployable, open-source framework that any food bank, public health department, or government agency can adapt to their region.